Automatically annotated response: (colored by assigned label)

```
["initializing"]Okay, so I came across this riddle. At first
glance, it seems tricky, but I can break it down.["end-section"]
["deduction"]Let me analyze each part separately. The first part:
"What has cities, but no houses." The answer must be something
that contains cities but lacks houses.["end-section"]
["adding-knowledge"]Cities have buildings, roads, and
infrastructure, but without houses, it must be something that
represents a city rather than an actual one.["end-section"]
["example-testing"]Maybe it's a city without residential
buildings, but that doesn't fit perfectly.["end-section"]
["backtracking"]Wait, perhaps it refers to something that
*displays* cities rather than being a real city.["end-section"]
["deduction"]What about the second part: "forests, but no
trees"?["end-section"]
["adding-knowledge"]Forests are made of trees, so if there are no
trees, it isn't really a forest.["end-section"]
["example-testing"]Unless it's an area that used to be a forest
but was deforested-but that wouldn't be called a forest
anymore.["end-section"]
["uncertainty-estimation"]That might be a stretch,
though.["end-section"]
["deduction"]The third part: "rivers, but no
water."["end-section"]
["adding-knowledge"]Rivers are defined by flowing water. If
there's no water, it might be referring to a representation of a
river rather than a real one.["end-section"]
["example-testing"]Maybe it's a dry riverbed, but then it's not
technically a river anymore.["end-section"]
["deduction"]Putting it all together, the answer must be something
that includes symbolic representations of cities, forests, and
rivers but lacks their real-world components.["end-section"]
["deduction"]A map fits this description-it has cities, forests,
and rivers marked on it, but they are not real.["end-section"]
```

C Cosine Similarity Between Feature Vectors

To better understand the relationships between different reasoning behaviors in thinking models, we analyze the cosine similarity between the extracted steering vectors for different behavioral categories. This analysis provides insights into how distinct these reasoning mechanisms are in the model's representational space and whether certain behaviors share similar underlying features.

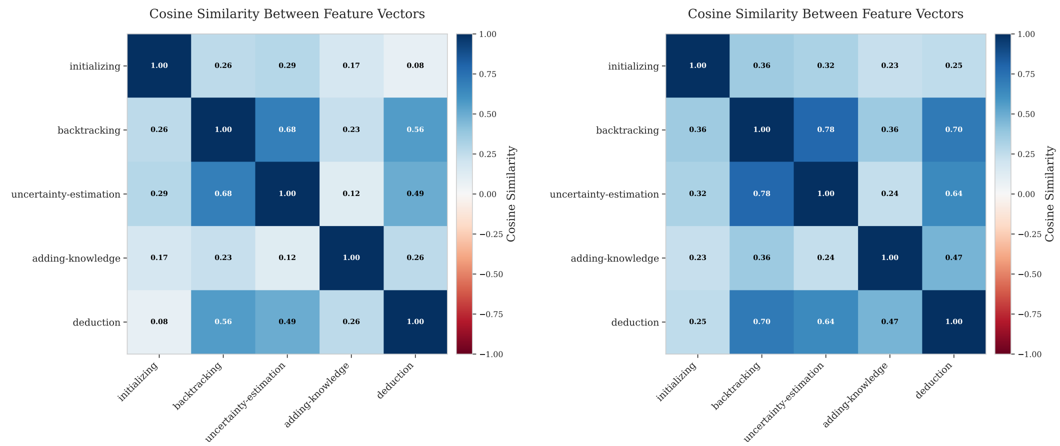
We compute pairwise cosine similarities between the steering vectors for five key behavioral categories: *initializing*, *backtracking*, *uncertainty-estimation*, *adding-knowledge*, and *deduction*. For each pair of behavioral categories, we calculate the cosine similarity between their corresponding steering vectors at the layers identified as causally relevant through attribution patching (Section 4.3).

The cosine similarity between two feature vectors $\mathbf{u}^{c_1}$ and $\mathbf{u}^{c_2}$ is computed as:

$$\text{sim}(\mathbf{u}^{c_1}, \mathbf{u}^{c_2}) = \frac{\mathbf{u}^{c_1} \cdot \mathbf{u}^{c_2}}{|\mathbf{u}^{c_1}||\mathbf{u}^{c_2}|}$$

Figure 5 presents the cosine similarity heatmaps for both DeepSeek-R1-Distill models. Several key observations emerge from this analysis:

1. *Distinct reasoning mechanisms*: Most behavioral categories show relatively low cosine similarities with each other, indicating that they correspond to distinct directions in the model's activation space. This supports our hypothesis that different reasoning behaviors are mediated by separate linear directions.



(a) DeepSeek-R1-Distill-Llama-8B

(b) DeepSeek-R1-Distill-Qwen-14B

Figure 5: Cosine similarity heatmaps between steering vectors for different reasoning behaviors. The heatmaps show pairwise similarities between feature vectors extracted for five behavioral categories. Values range from -1 (completely opposite) to 1 (identical direction), with colors indicating the strength and direction of similarity. Most behaviors show low to moderate similarities, indicating they represent distinct reasoning mechanisms in the model's activation space.

2. *Uncertainty and backtracking correlation*: Both models show moderate positive correlation between *uncertainty-estimation* and *backtracking* behaviors, which aligns with our intuition that models often express uncertainty before deciding to change their reasoning approach. Despite this cosine similarity overlap, our steering experiments (see Section 4.5) demonstrate that these represent fundamentally different mechanisms in the model's reasoning process.
3. *Model-specific patterns*: While the overall structure is similar across models, there are notable differences in the specific similarity values, suggesting that different model architectures may encode these reasoning behaviors with varying degrees of separation.

These results demonstrate that the extracted steering vectors capture meaningful and largely orthogonal directions in the model's representational space, validating our approach for isolating specific reasoning mechanisms in thinking language models.

D Cosine Similarity with Embedding and Unembedding Vectors

To better understand the relationship between our extracted steering vectors and the model's embedding space, we analyze the cosine similarity between steering vectors and the model's embedding and unembedding matrices across different layers. This analysis helps explain why certain layers are more effective for steering than others.

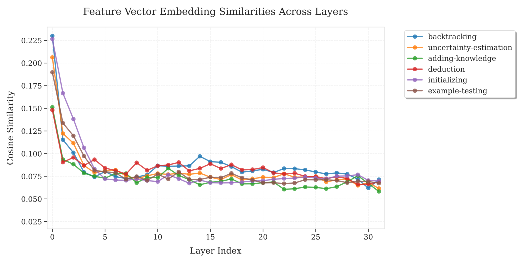
D.1 Methodology

We compute the cosine similarity between each steering vector at each layer and the model's embedding matrix (`model.embed_tokens.weight`) and unembedding matrix (`model.lm_head.weight`). For each steering vector $\mathbf{u}_\ell^c$ at layer ℓ and behavioral category c, we calculate:

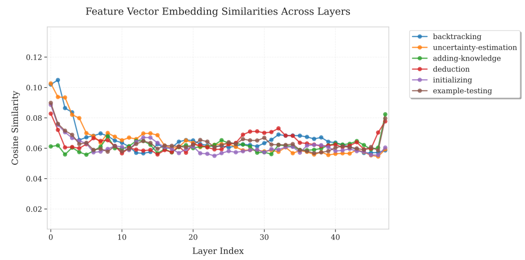
$$\text{sim}_{\text{embed}}(\mathbf{u}_\ell^c) = \max_i \cos(\mathbf{u}_\ell^c, \mathbf{E}_i) \quad (1)$$

$$\text{sim}_{\text{unembed}}(\mathbf{u}_\ell^c) = \max_j \cos(\mathbf{u}_\ell^c, \mathbf{U}_j) \quad (2)$$

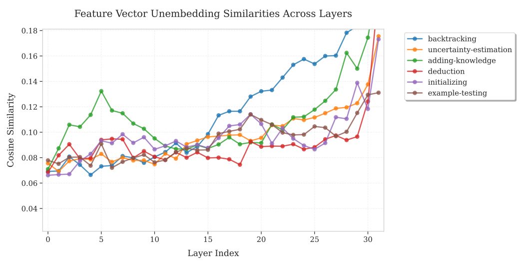
where $\mathbf{E}_i$ represents the i-th row of the embedding matrix and $\mathbf{U}_j$ represents the j-th row of the unembedding matrix.



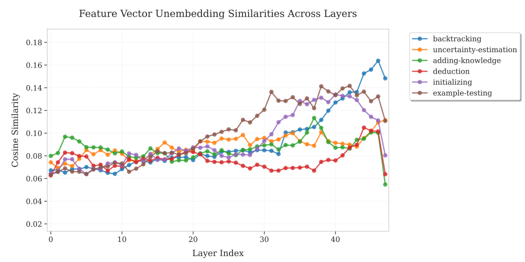
(a) Embedding similarities - Llama-8B



(b) Embedding similarities - Qwen-14B



(c) Unembedding similarities - Llama-8B



(d) Unembedding similarities - Qwen-14B

Figure 6: Layer-wise cosine similarities between steering vectors and embedding/unembedding matrices. The Llama model shows high embedding similarity in early layers, indicating strong correlation with token representations. This pattern is not observed in the Qwen model, explaining the different layer selection strategies for effective steering.

D.2 Results and Analysis

Figure 6 shows the layer-wise similarity patterns for both models. The analysis reveals distinct architectural differences:

DeepSeek-R1-Distill-Llama-8B: Shows high similarity with embedding vectors in the first layers, explaining the high KL divergence observed at the beginning of the attribution plots (Figure 3). This suggests that Llama retains token representation information in early layers, making steering vectors in these layers too correlated with specific tokens to be effective for behavioral control.

DeepSeek-R1-Distill-Qwen-14B: Does not exhibit this pattern, with more uniform similarities across layers. This architectural difference allows for more flexibility in layer selection for steering.

These findings inform our layer selection strategy: we prioritize layers with high KL divergence in attribution plots while avoiding early layers that show excessive correlation with embedding tokens. This approach ensures that our steering vectors capture behavioral patterns rather than token-specific representations.

E Steered Example (Adding Knowledge)

Task:

```
What's the probability of drawing exactly two hearts when dealing
three cards from a standard deck?
```

Original Response:

```
["initializing"]
Okay, so I need to figure out the probability of drawing exactly
two hearts when dealing three cards from a standard deck.
["end-section"]
```